

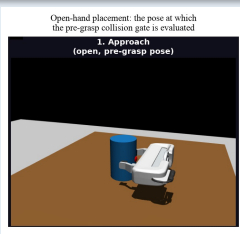
Counterfactual Diagnosis of Robotic Grasp Failures under Perceptual and Geometric Degradation

MuJoCo · Contact-GraspNet · 7,560 physically scored trials · simulator-backed counterfactual ground truth

Problem & research question

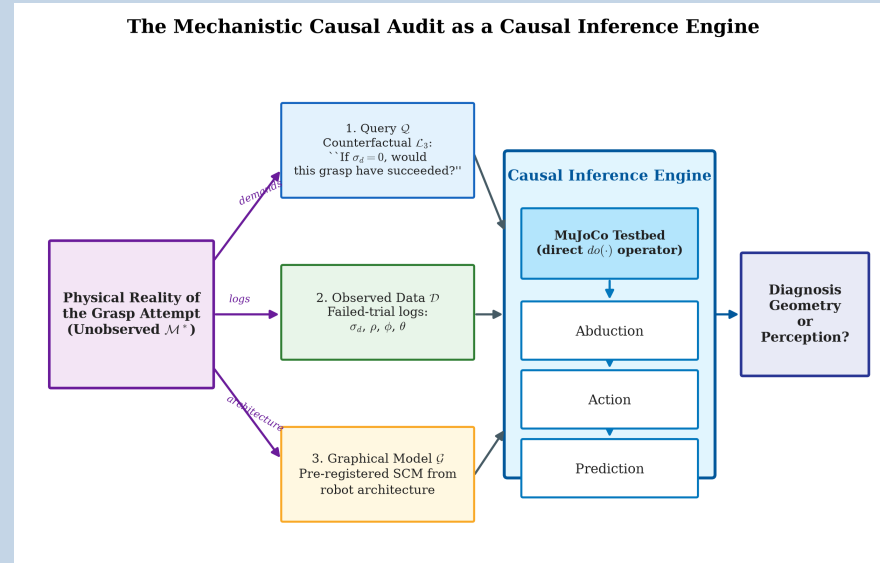
A failed grasp reveals the outcome—but not whether the root cause was sensing, viewpoint, object geometry, or execution.

Can a mechanistic SCM identify which controlled factor caused this specific failure—and recognise when no single fix is sufficient?



- Contact-GraspNet proposes a top-ranked 6-DoF pose.
- The open hand may already intersect the object or table.
- A fluent post-hoc explanation cannot verify its own cause.

Method: a simulator-backed causal inference engine



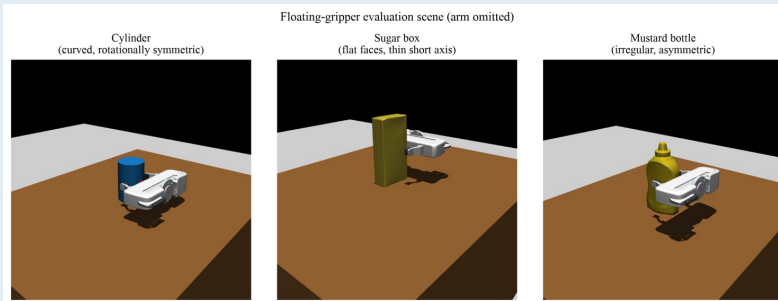
What is controlled?

- Depth noise σ_d
- Point retention ρ
- Elevation ϕ and azimuth θ

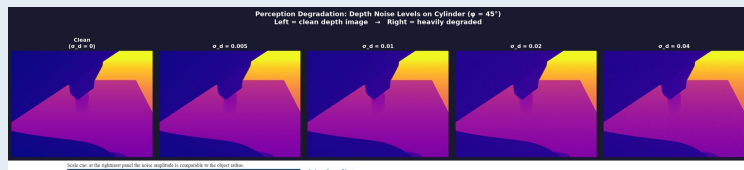
What makes it causal?

- Graph pre-registered from known dataflow
- Same seed replayed; one variable changed
- Diagnosis checked by re-simulation

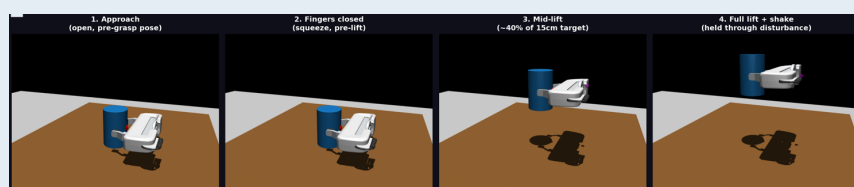
Physical experiment & success criterion



Three geometries expose different contact and pose-selection failure modes.



Same simulated scene, progressively corrupted depth.



- 7,560 trials: 3 objects $\times$ 7 $\sigma_d \times$ 2 $\rho \times$ 6 $\phi \times$ 3 $\theta \times$ 5 seeds.
- Unfiltered top-1 pose; collision gate evaluated with the hand open.
- Physical outcome: close, lift 15 cm, and hold through a shake.
- Three-stage factorisation separates proposal, geometry, and execution.

Physical result: the geometric gate

CONFIRMATORY · PHYSICAL Y

5.0%

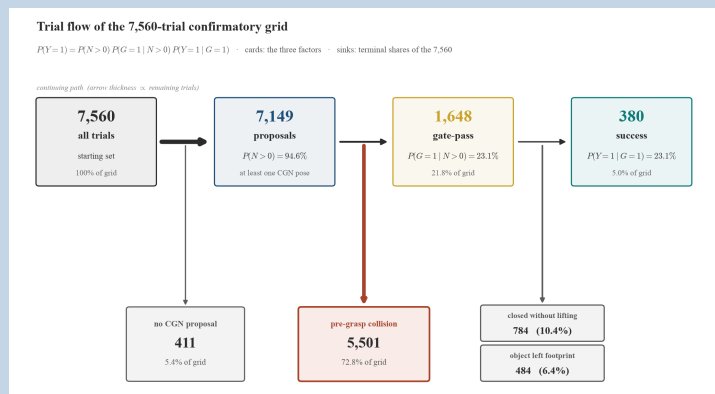
marginal success

72.8%

pre-grasp collision

23.1%

hold | gate-pass

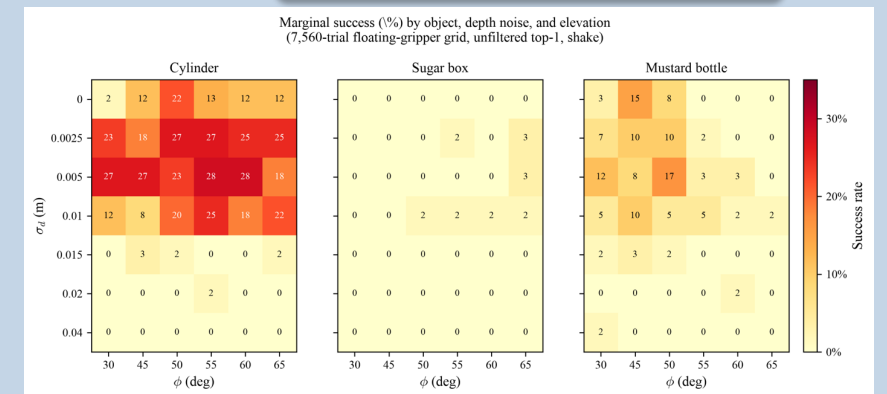


Depth noise moves the two mechanisms in opposite directions:

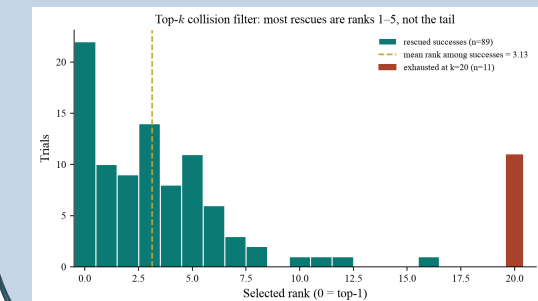
- More noise makes poses drift away, so fewer collide at the open-hand gate.
- But among poses that clear the gate, physical hold collapses sharply.
- The pooled success rate hides this geometric/execution reversal.

Object geometry dominates

CONFIRMATORY · PHYSICAL Y



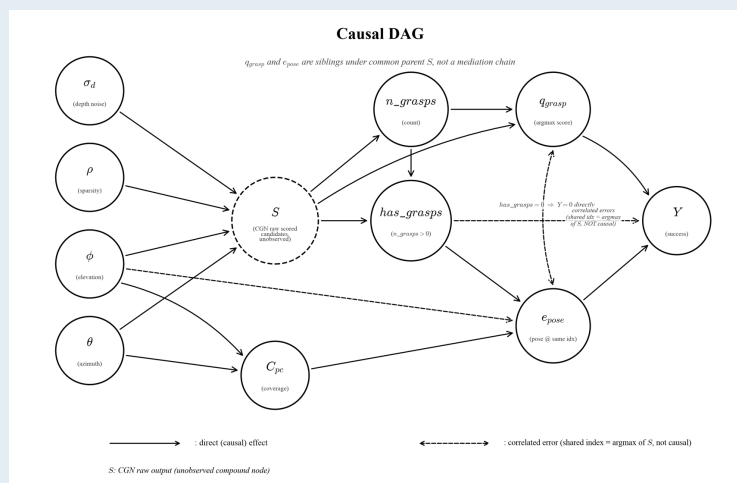
Post-gate success: box 1.5% · cylinder 35.4% · mustard 35.0%



89%

top-20 collision-filtered success at one favourable clean cell
Ceiling check only; post-hoc viewpoint and lighter clamp-and-lift criterion.

Counterfactual diagnosis



1. Abduction

Recover the realised noise terms U .

2. Action

Replace one equation with $do(X=x')$.

3. Prediction

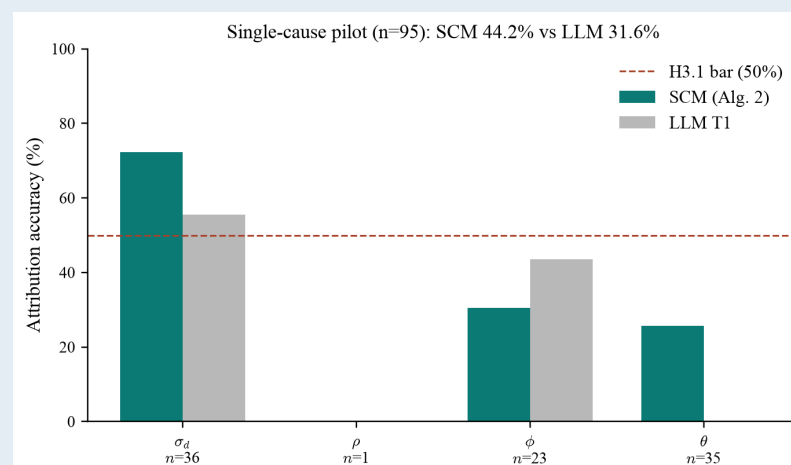
Propagate forward to counterfactual Yx' .

A failure is attributed only when the single reset changes the predicted outcome; otherwise it remains joint or irreducible within the tested reset.

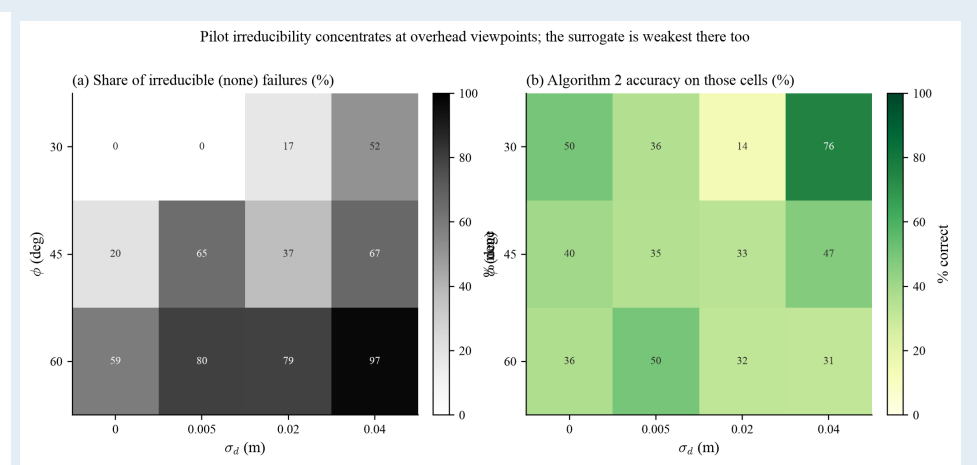
Pilot diagnosis: SCM vs LLM

Scored on the proximity outcome—not physical grasp success

PILOT · PROXIMITY Y



SCM: 42/95 = 44.2% | LLM: 30/95 = 31.6% | pre-registered 50% bar missed



57.2% of 292 pilot failures have no single-variable fix; irreducibility concentrates at overhead viewpoints.

Take-home message

Failure is not one mechanism. In the physical grid, top-ranked pose selection creates a dominant geometric gate, depth noise drives the gate and execution in opposite directions, and object geometry overwhelms viewpoint effects.

The mechanistic audit makes those layers explicit and tests diagnoses by rewinding the same trial. It improves on the LLM in the pilot, but does not meet the pre-registered 50% accuracy target.